

Zekun Wu

[EMAIL](#)[WEBSITE](#)[LINKEDIN](#)

EDUCATION

Saarland University*Ph.D. in Computer Science*

Saarbrücken, Saarland, Germany

*Feb. 2023 – Present***Washington University in St. Louis***Master of Science in Data Analytics and Statistics*

St. Louis, MO, USA

*Sep. 2019 – May 2021***Beihang University***Master of Science in Control Science and Engineering*

Beijing, China

*Sep. 2016 – Jan. 2019***Nanjing University of Aeronautics and Astronautics***Bachelor of Engineering in Mechanical Engineering and Automation*

Nanjing, Jiangsu, China

Sep. 2008 – July 2012

PUBLICATIONS

1. **Z. Wu**, X. Wang, R. Y. Pang, C. Wang and A. M. Feit (2026). From Review to Reuse: How Post-Task Workflow Can Support Human-AI Agent Interaction. *ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*. (under review) [\[details\]](#)
2. **Z. Wu**, M. Su, H. Li, T. Nagashima and A. Feit (2026). Gaze-Informed Proactive AI Assistance for Children's Picture Exploration. *ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*. (under review) [\[details\]](#)
3. **Z. Wu**, M. Jobanputra, V. Demberg, J. Hullman and A. Feit (2026). How Information Detail and Confidence in AI Responses Shape User Beliefs. *The 2nd International Conference on Human-AI Interaction and Experience Design (HAXD26)*. [\[details\]](#)
4. **Z. Wu** and A. Feit (2025). Understanding and Predicting Temporal Visual Attention Influenced by Dynamic Highlights in Monitoring Task. *IEEE Transactions on Human-Machine Systems (IHMS)* [\[details\]](#).
5. **Z. Wu**, Y. Wang, M. Langer and A. Feit (2025). RelEYEance: Gaze-based assessment of users' AI-reliance at run-time. *ACM Symposium on Eye Tracking Research and Application (ETRA 2025)*. [\[details\]](#)
6. **Z. Wu** and A. Feit (2024). Enhancing Saliency Prediction in Monitoring Tasks: The Role of Visual Highlights. *ACM Symposium on Eye Tracking Research and Application (ETRA 2024)*. [\[details\]](#)
7. A. Das*, **Z. Wu***, I. Škrjanec, and A. Feit (2023). Shifting Focus with HCEye: Exploring the Dynamics of Visual Highlighting and Cognitive Load on User Attention and Saliency Prediction. *ACM Symposium on Eye Tracking Research and Application (ETRA 2024)*. [\[details\]](#)
8. S. Doroudian, **Z. Wu**, A. Galati, W. Wang, and A. Lu (2021). A Study of Real-time Information and Immersive Maps on User Behaviors during Search and Rescue (SAR) Training of Firefighters. *IEEE VR 2022 Workshop on 3D Content Creation for Simulation Training (TrainingXR)*. [\[details\]](#)
9. K. Wu and **Z. Wu** (2020). A Human-Centered Risk Model for Construction Safety. *IEEE Access*. [\[details\]](#)
10. **Z. Wu**, P. Xin, X. Zhao, and Y. Jiang (2019). The Task Demands-Resources Method: A New Approach to Human Reliability Analysis from a Psychological Perspective. *Quality and Reliability Engineering International*. [\[details\]](#)
11. X. Pan and **Z. Wu** (2018). Performance Shaping Factors in Human Error Probability Modification of Human Reliability Analysis. *International Journal of Occupational Safety and Ergonomics*. [\[details\]](#)

12. **Z. Wu**, X. Pan, and X. Chen (2017). Relation of Motivation Intensity, Stress Level, and Human Performance: A Human Reliability Experiment. *Proceedings of the 11th Asian Control Conference (ASCC 2017)*, Gold Coast, 2017. [\[details\]](#)
13. **Z. Wu**, X. Pan, H. Wang, and J. Chen (2017). Influence of Work Motivation and Task Difficulty on Human Reliability. *Proceedings of the 2nd International Conference on Reliability Systems Engineering (ICRSE 2017)*, Beijing, 2017. [\[details\]](#)

PROJECTS

Trace2Flow: Post-Task Workflows for AI Agents [\[demo\]](#) April 2026 – Present

- Developed Trace2Flow, a research prototype that converts AI-agent execution traces into editable workflows for inspecting, validating, and reusing multi-step tasks.
- Designed an LLM-based workflow generation pipeline and evaluated it with 20 participants, showing improved understanding and error detection compared with prompt-only review.

Gaze-Based Proactive AI Assistant for Children in Picture Exploration [\[demo\]](#) June 2025 – April 2026

- Designed a gaze-driven framework that detects children’s attentional states during reading activities using eye-tracking data.
- Developed proactive assistance strategies powered by large language models (LLMs) to automatically select the areas that children interested in picture.

Examining AI Prompt Design for Influencing User Beliefs Jan 2025 – Present

- Designed and ran a large-scale online experiment to evaluate how AI response detail and confidence affect user belief changes in factual and opinion tasks.
- Developed a belief measurement framework and statistical models that uncovered how specific AI message features can systematically shift user beliefs.

Temporal Saliency Analysis with Visual Highlighting Feb 2023 – Present

- Developed a multiple drone monitoring interface prototype, examining the effects of visual highlighting on user gaze behavior.
- Collected a gaze dataset to document variations in user gaze patterns influenced by visual highlights, facilitating detailed user behavior analysis and interface optimization.
- Employed deep-learning-based saliency models to predict user fixation patterns, aiding in the comprehension of user-interactive interface effectiveness.

Future of Firefighting and Career Training Sep 2021 – Feb 2023

- Handled data collection and analysis in an immersive virtual reality simulation system, focusing on firefighting search and rescue scenarios.
- Developed a model to estimate user stress levels using a stacked-LSTM layer neural network.

EXPERIENCE

Researcher, E6 Project (CPEC Group) Feb. 2023 – Present *Saarland University* *Saarbrücken, Saarland, Germany*

- Led research on eye-tracking in human–AI interaction, spanning visual attention prediction, real-time inference of user reliance on AI, and gaze-driven proactive assistance from LLM, combining experimental design with advanced computational modeling.
- Engaged in international research collaborations, enriching the project’s depth and scope; actively contributed to collaborative meetings and knowledge exchange sessions.
- Organized seminars, facilitated student discussions on emerging models, and supervised master students.

Graduate Research Assistant Sep. 2021 – Feb. 2023 *UNC Charlotte* *Charlotte, NC, USA*

- Designed and implemented a firefighter search and rescue experiment utilizing a VR system, focusing on realistic simulation and data accumulation.

- Managed the collection, processing, and modeling of data generated from the VR experiment, paving the way for more immersive training experiences.

Assistant in Instruction

Jan. 2020 – May. 2021

Washington University in St. Louis

St. Louis, MO, USA

- Conducted 8 comprehensive seminar lectures on R programming, facilitating learning for 50+ undergraduate students.
- Offered office hours to review material, address questions, and assist students with assignments, enhancing their understanding and performance.
- Tasked with the development and grading of assignments, quizzes, and exams, ensuring accurate evaluation of student understanding and progress.

Graduate Research Assistant

Sep. 2016 – Mar. 2019

Beihang University

Beijing, China

- Completed Master's research focuses on human reliability analysis during the fueling process of spacecraft launch activities.
- Designed a substantial simulation program for rocket propellant fueling operations
- Led the development and publications, contributing valuable insights to the field.

Junior Maintenance Engineer

Jul. 2012 – Jun. 2015

SF Airlines

Shenzhen, Guangdong, China

- Crafted compliance bulletins for frontline engineers, interpreting manufacturers' maintenance manuals with accurate translations and documentation.
- Offered technical support to maintenance technicians, aiding in diagnosing, troubleshooting, and documenting issues associated with aircraft engines.

SKILLS AND COMPETENCIES

Programming Languages: Python (pytorch, scikit-learn, numpy, pandas), Java, SQL, JavaScript, HTML/CSS, R

Large Language Model Tools: OpenAI API, LangChain, Hugging Face Transformers

Software: SPSS, SAS

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Eclipse

Mathematics: Estimation and Detection Theory, Matrix Theory, Numerical Analysis, Optimization, Probability and Stochastic Process